# Analytic Multi-Loop Helicity Amplitudes in QCD

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The full physical information of a color-ordered helicity amplitude is in the finite remainder:

$$\mathcal{R}^{\ell-\text{loop}}_{n-\text{point}}(|i\rangle_\alpha, [i|_{\dot\alpha}) = \sum_i r_i(|i\rangle_\alpha, [i|_{\dot\alpha})\, G_i(|i\rangle_\alpha \times [i|_{\dot\alpha})\,.$$

Catani ('98)
Becher, Neubert ('09)
Gardi, Magnea ('09)
Weinzierl ('11)

***Challenges:***

- Compute the $G_i$ , a basis of transcendental functions, e.g. the *Pentagon Functions*;

  Chicherin, Sotnikov ('20) + Zoia ('21) + Abreu, Ita, Page, Tschernow ('23)

- Reconstruct the $r_i$ , functions in the field of fractions over a polynomial quotient ring,

  Peraro ('16); GDL, Maitre ('19); GDL, Page ('22).

$$R_n = \mathbb{F}\Big[|1\rangle_\alpha, [1|_{\dot\alpha}, \ldots, |n\rangle_\alpha, [n|_{\dot\alpha}\Big]\Big/ J \qquad \text{with} \qquad J = \text{Ideal}\Big(\textstyle\sum_{i=1}^n |i\rangle[i|\Big)\,,$$

in connection with algebraic geometry, and extended to massive particles (spin-spinors).

Arkani-Hamed, Huang, Huang ('17); Ochirov ('18); Shadmi, Weiss ('18); Campbell, GDL, Ellis ('25); GDL, Melnikov, Tresoldi ('26)

***Amplitudes we computed:*** GDL + [Abreu, Campbell, Ellis, Ita, Klinkert, Kuschke, Maitre, Page, Ruf, Seth, Sotnikov, ...]

| Inspire | Process | Loops | Year | Inspire | Process | Loops | Year | Inspire | Process | Loops | Year |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1728738 | $0 \to ggggg$ | 1L FC | 2019 | 2060823 | $0 \to q\bar{q}\ell\bar{\ell}\ell'\bar{\ell}'g$ | 1L FC | 2022 | 2915977 | $q\bar{q} \to t\bar{t}H$ | 1L FC | 2025 |
| 1779473 | $pp \to Hjj$ | 1L FC | 2020 | 2663067 | $pp \to \gamma\gamma\gamma$ | 2L FC | 2023 | 2954479 | $q\bar{q} \to HHH$ | 1L FC | 2025 |
| 1826610 | $pp \to jjj$ | 2L LC | 2020 | 2723256 | $pp \to jjj$ | 2L FC | 2023 | 3128446 | $pp \to Vjj$ | 2L LC | 2026 |
| 1861752 | $pp \to W(\ell\nu)\gamma j$ | 1L FC | 2021 | 2821186 | $pp \to HHj$ | 1L FC | 2024 | 3152032 | $pp \to Hjj$ | 2L LC | 2026 |
| 2048108 | $pp \to \gamma\gamma\gamma$ | 2L LC | 2022 | 2899891 | $pp \to V(\ell\bar{\ell})jj$ | 2L LC | 2025 | | | | |

Least common denominator degrees of freedom $pp \to \gamma\gamma\gamma$ 30k; $jjj$ 250k ; $Vjj$ 25M; $Hjj$ 80M.